Moon reading light v0.1: deliverable pack

This folder holds everything needed to manufacture, print, flash, assemble and test the lamp, in build order. It is **generated** by `python3 tools/build_deliverable.py` from commit `67f9752`. Don't edit it by hand: change the design, then re-run the generator. `MANIFEST.txt` lists every file with its SHA-256 and its source in the repo.

Nothing has been ordered. Matt places every order and starts every print. The design has been through two verification rounds; it has not been built or tested on hardware yet.

Build order

Step	Folder	What you do	Instructions
0	`00-overview/`	Read the plan, look at the renders, check the open decisions	`00-START-HERE.pdf`, `PLAN-…pdf`
1	`01-pcb-manufacture/`	Upload each board's Gerber zip to PCBWay (or JLCPCB) with the specs in its `FAB_NOTES`	`handoff-01-pcb-order.pdf`
2	`02-parts-order/`	Paste `farnell_quickpaste.txt` into Farnell's Quick Paste; buy the off-board items	`handoff-02-parts-order.pdf`
3	`03-3d-printing/`	Print the coupons first, then the moon head, the pack and the jig	`handoff-03-printing.pdf`
4	`04-firmware/`	Set the fuses once, then flash `release/` (or `debug/` for bring-up)	`handoff-04-firmware.pdf`
5	`05-assembly-and-bring-up/`	Assemble the boards, cable, moon and pack; first light	`handoff-05-assembly-bringup.pdf`
6	`06-test-and-verification/`	Bench checklist: electrical, firmware, thermal, optical, fit	`VERIFY-bench-checklist.pdf`
7	`07-design-reference/`	Background: electronics notes, datasheet specs, research	—

Status of the files in this pack

Board	ERC	DRC errors	DRC warnings	Unconnected	Schematic parity
main-board	0	0	2	0	0
led-board	0	0	0	0	0

The main board's 2 DRC warnings are intended: the slide-switch outline crosses the board edge where its knob overhangs. Every printed part passes the 3dprint-contract checks (11 parts). The firmware's 29 host tests pass, and all four builds compile without warnings.

Decisions still open for Matt

See `00-overview/open-decisions-and-round1-summary.pdf` §5. In short:

1. Reading brightness is 20–30 lux on the page, which suits a moon lamp, not a full reading light. v0.2 option: a downward reading LED.
2. No cell-temperature sensing on the charger, so charge at room temperature. v0.2 option: an NTC charger.
3. Soft-orange night mode includes some warm white (recommended: keep it).

4. Green/blue drift at low battery (accepted for v0.1).

Safety rules that apply to every step

- Meter the battery lead before plugging it in: + goes to J2 pin 1 (silk "+").
- Never power the board from a programmer or USB-UART: J4 pin 1 is a sense line only.
- Charge at room temperature, and stop using a pouch that swells.
- Do not use anaerobic thread-locker on the printed parts (it crazes PETG).